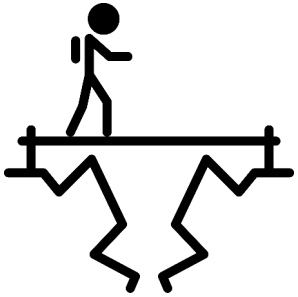
Challenges with Anomaly Detection

What are we going to learn?



Challenges with an Anomaly Detection Application

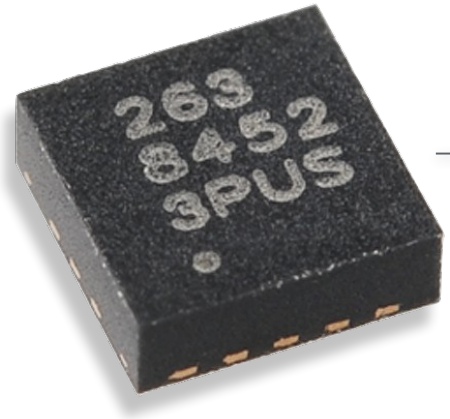
Anomaly Detection **ML Pipeline**

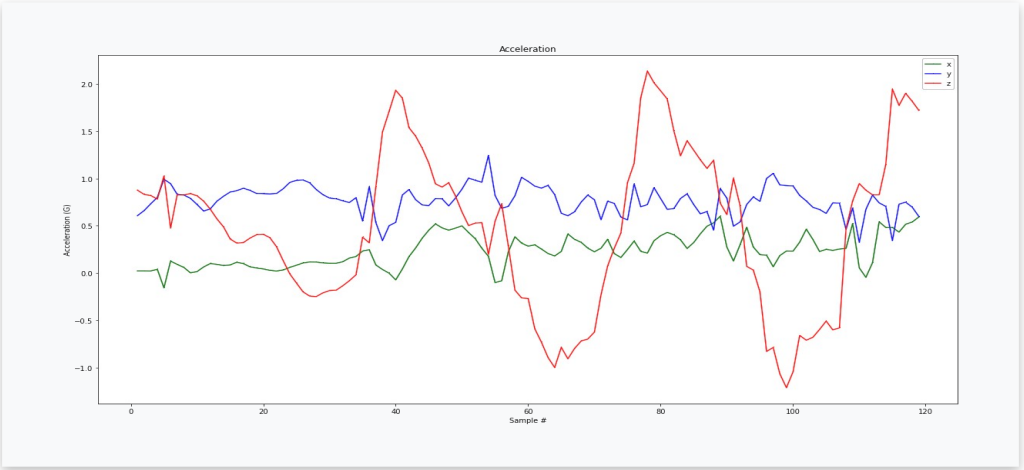
Training, Testing in Colab

Application: Factory machinery



Sensor: **Accelerometer**





Constraints for **on-device computing**

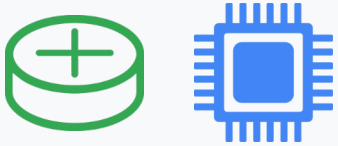


Latency

Constraints for **on-device computing**



Latency

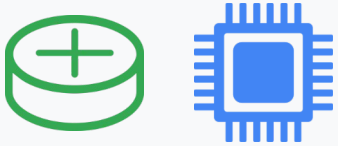


Limited Devices

Constraints for **on-device computing**



Latency



Limited Devices



Accuracy

Constraints for **on-device computing**

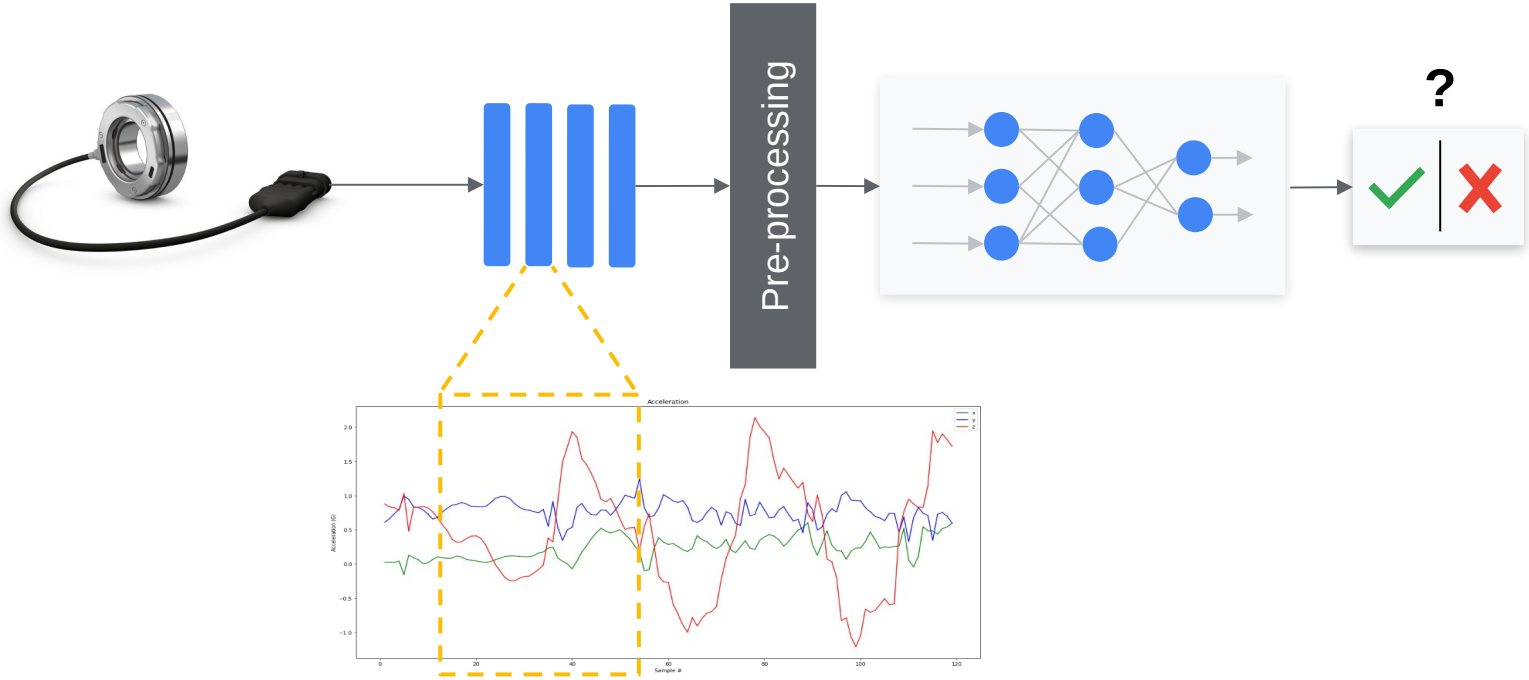


Latency

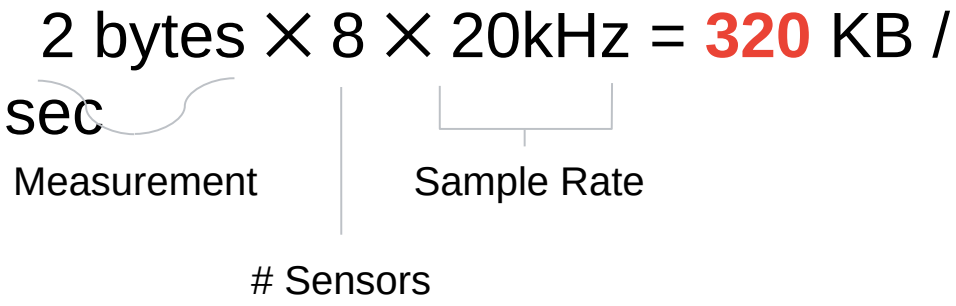
Limited Devices

Accuracy

Real Time Constraint



It’s **too expensive** to stream to the cloud

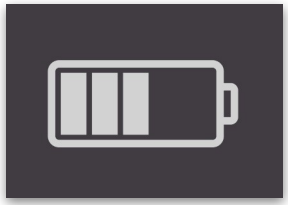


It’s **too expensive** to stream to the cloud

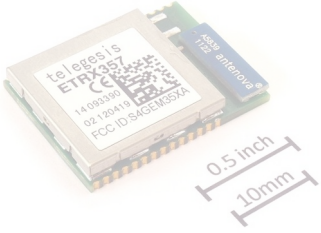
2 bytes ✕ 8 ✕ 20kHz = **320** KB / sec



It’s **too expensive** to stream to the cloud



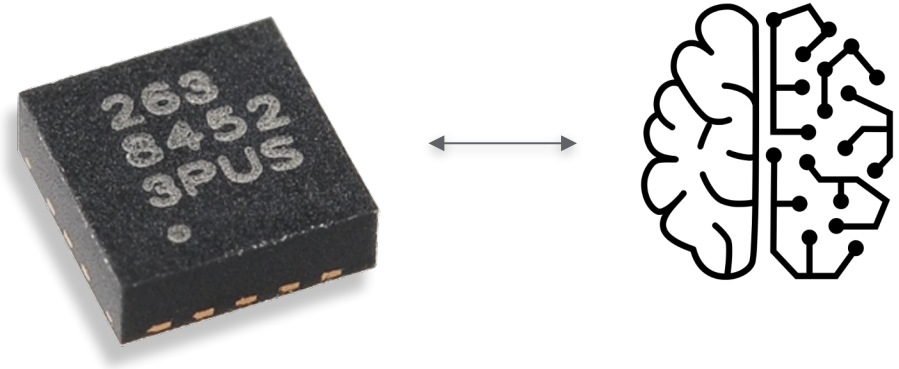
2 bytes ✕ 8 ✕ 20kHz = **320** KB / sec



30 KB / sec

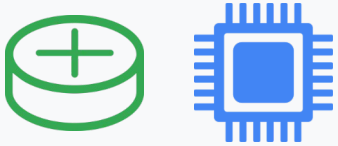
30 KB/s MAX

Need “**intelligence**” *close* **to sensors**



Constraints for **on-device computing**

Latency

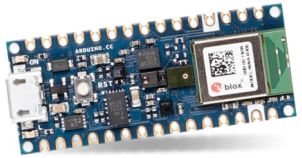


Limited Devices

Accuracy

Very Small (Tiny) Devices

- Cost ($) ✔
- Power (W) ✔
- Eng. Effort ✔



Our board [Course 3 Kit] only has **256KB** of RAM (memory)

Constraints for **on-device computing**

Latency

Limited Devices

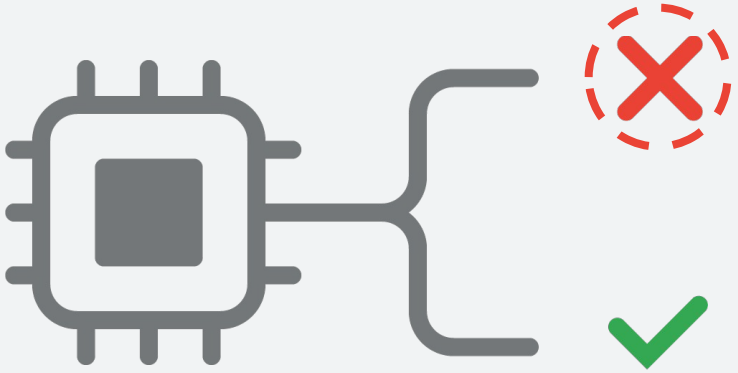


Accuracy

False Negative

Catastrophic impact

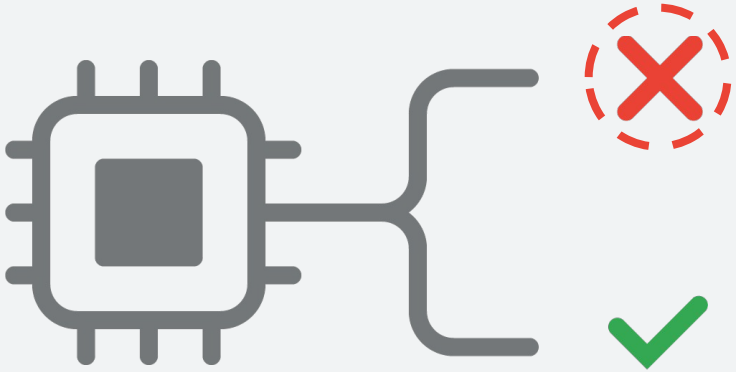




False Negative

Catastrophic impact





REAL

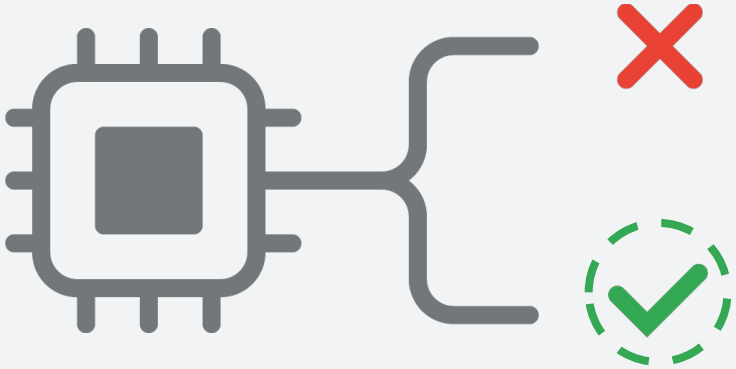
ERROR

False Positive

False alarm, **cost** impact





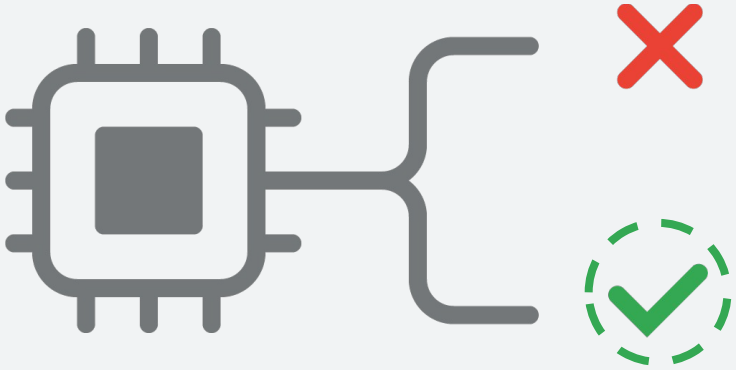


False Positive

False alarm, **cost** impact









ERROR

Fullscreen
Show Presenter